Decentralized Edge Workload Forecasting Using Semantic Gossip Learning and Agentic AI

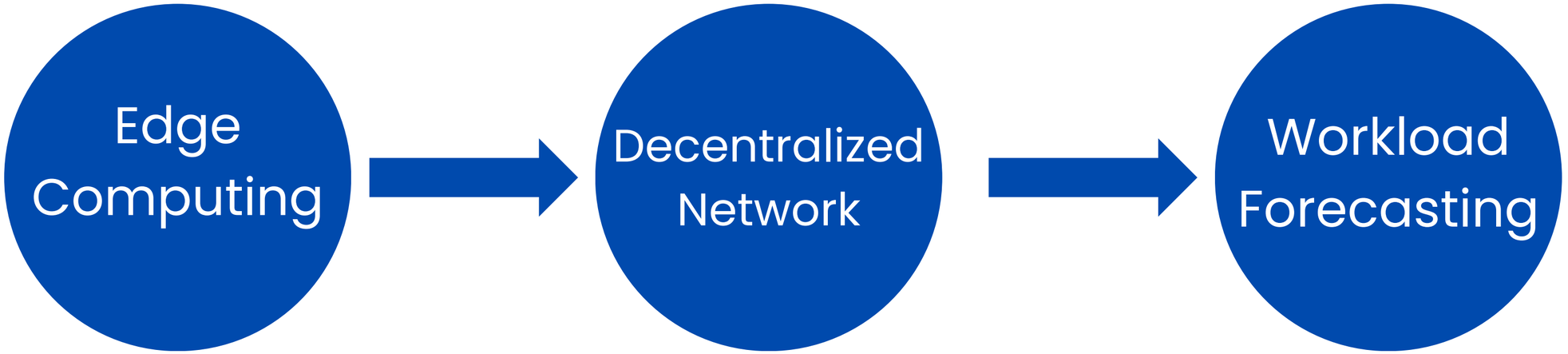
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Introduction



Edge computing brings computation closer to data sources, enabling low-latency processing for distributed applications.

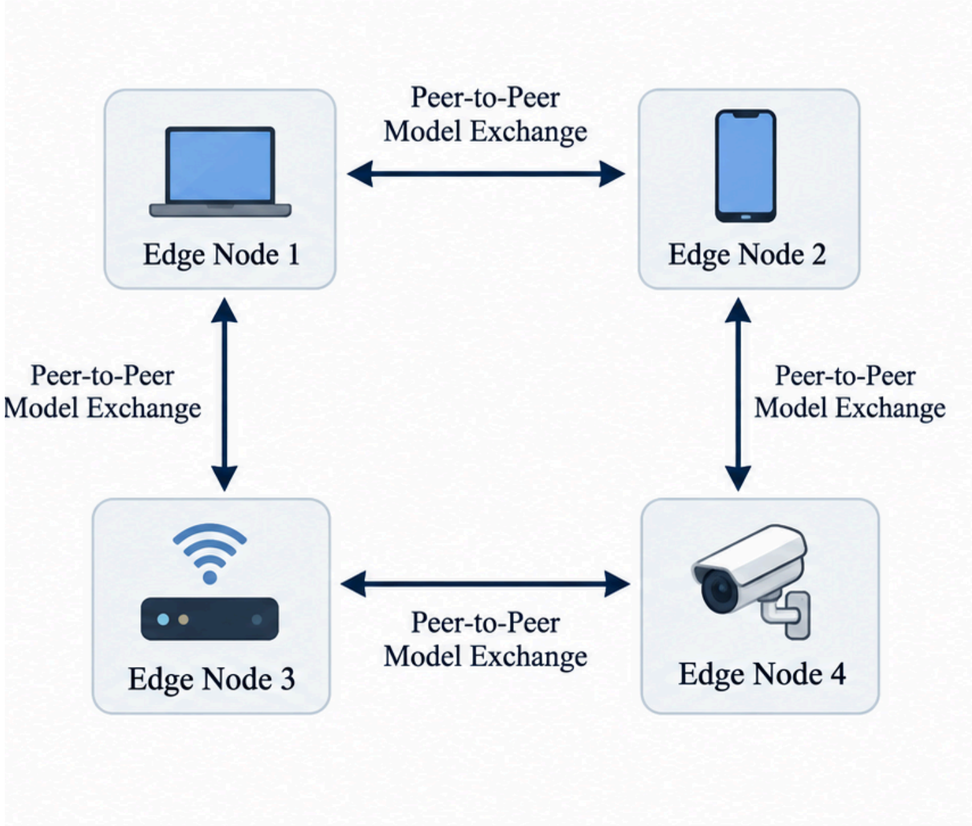
As edge environments become distributed, decentralized coordination can allow multiple edge nodes to cooperate without relying on a central coordinator.

Forecasting workload at the edge can help anticipate resource demand and support efficient resource management.

Motivation/Background

Why Gossip Learning for Decentralized Edge Networks?

1. **No Central Coordinator**
2. **Local Peer-to-Peer Communication**
3. **Suitable for Distributed Edge Environments**



Key Motivation: Gossip Learning provides a decentralized way for edge nodes to collaboratively exchange learned information without relying on a central coordinator.

Literature Review

Evolution toward decentralized workload intelligence

Sr.No	Study Reference	Key Insight
1	Ciavotta et al. (2021)	Establishes a decentralized serverless edge architecture without a central coordinator.
2	Tundo et al. (2025)	Demonstrates decentralized workload forecasting using asynchronous Gossip Learning and addresses convergence, communication overhead, and resilience.
3	Yao et al. (2025)	Demonstrates MARL-based adaptive resource orchestration for autonomous coordination in dynamic environments.

Research Gap

1. The "Blind" Communication Problem (Tundo et al.)

- Model updates are not explicitly filtered by their predictive value.
- Redundant updates increase communication and energy overhead.

2. Passive Forecasting

- Existing work focuses mainly on workload prediction
- Forecasts are not directly coupled with autonomous resource decisions.

Research Opportunity: Communication-Aware Forecasting → Autonomous Edge Decisions

Objectives



Design the Adaptive Predictive-Semantic Model (APSM) to actively suppress model transmissions unless the local node encounters a statistically significant "surprise" in traffic.

Deploy Multi-Agent Reinforcement Learning (MARL) on edge nodes. Autonomous agents will ingest the APSM forecasts and negotiate resource offloading in real-time

Validate the stability of both the APSM filter and Agentic AI on massive, large-scale (50-node and 100-node) edge topologies.

Research Gap

1. The "Blind" Communication Problem (Tundo et al.)

- Gossip Learning improves decentralized forecasting, but model transmission is not explicitly guided by the predictive value of each update.
- It treats all data as equal. Nodes waste severe bandwidth and energy transmitting highly predictable, "boring" traffic patterns.

2. Passive Forecasting

- Current edge literature focuses heavily on generating accurate workload forecasts, but stops there.
- The Flaw: A forecast is useless if the system doesn't automatically act on it. There is a lack of autonomous, proactive task offloading based on these predictions.

Methodology: Local Workload Forecasting

The Input Data (Time-Series)

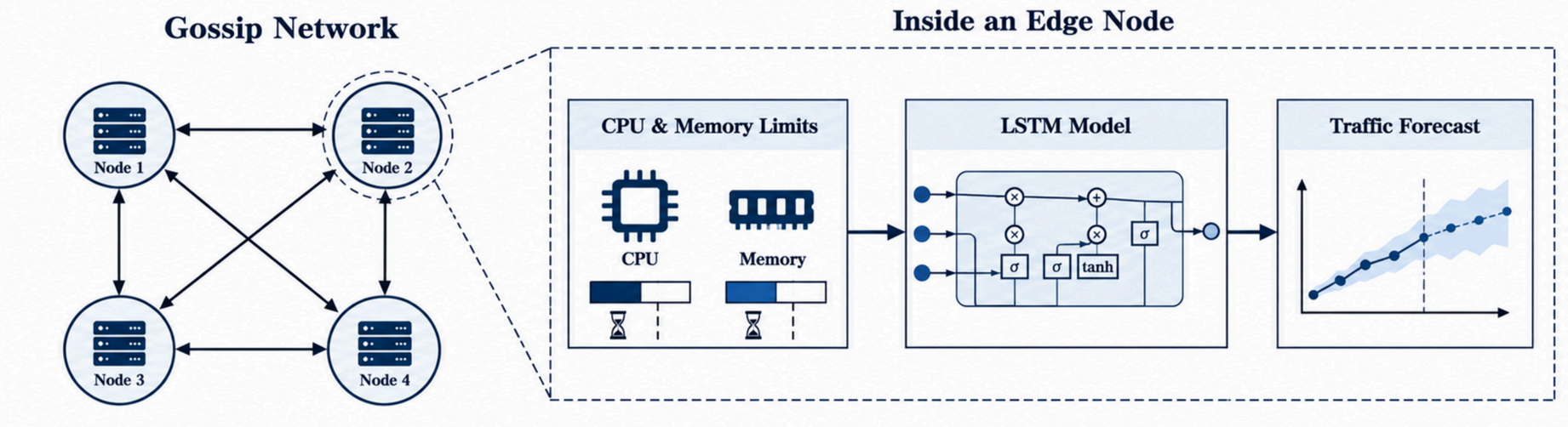
- Each node continuously ingests real-world, time-series workload data.
- Dataset used: Microsoft Azure Serverless function traces mapped to Porto taxi spatial mobility data

Local Forecasting (LSTM)

- Due to strict edge hardware limits (CPU/Memory), nodes run a highly lightweight LSTM Neural Network (1 hidden layer, 16 units).
- The LSTM learns continuously in real-time to predict the upcoming workload traffic

Network Communication (Gossip Learning)

- Instead of sending this raw data to a central server, the node packages its newly trained LSTM weights and prepares to share its "knowledge" with its peers over the Gossip Network.



Methodology: The APSM Filter

1. Absolute Prediction Error

$$\epsilon(t) = |y(t) - \hat{y}(t)|$$

- Calculates the "Semantic Surprise" of local traffic

2. Adaptive Threshold

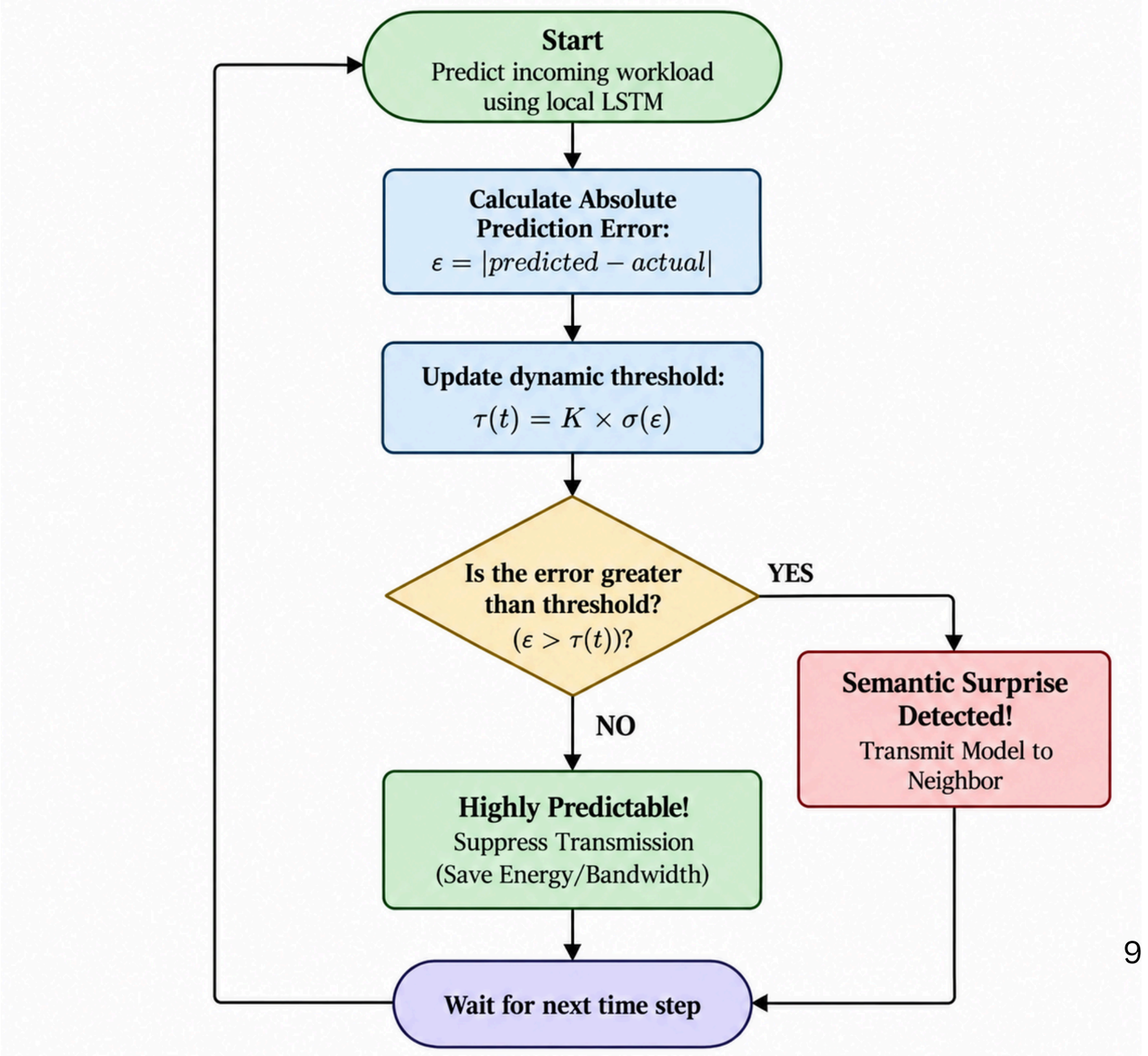
$$\tau(t) = K \times \sigma(\epsilon)$$

- Dynamically scales based on rolling standard deviation

3. The Semantic Suppression Rule

$\epsilon(t) > \tau(t) \rightarrow$ Transmit Model

$\epsilon(t) \leq \tau(t) \rightarrow$ Suppress Model

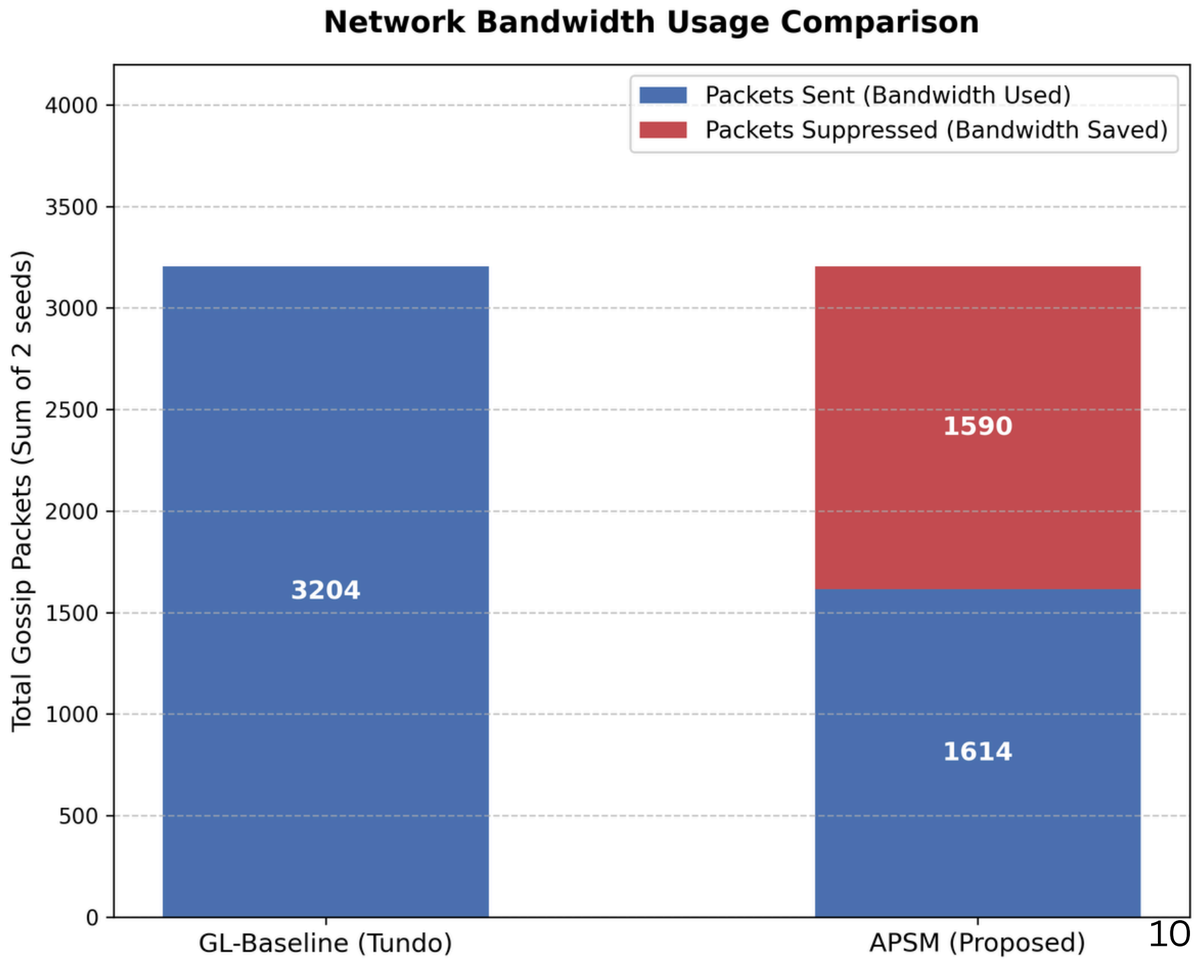


Results

1. PERFORMANCE COMPARISON

Metric	Gossip Learning	APSM Filter
Total Packets Sent	3,204	1,614
Packets Suppressed	0	1,590
Bandwidth Reduction	0%	49.61%
Avg. Scaled MSE	0.0318	0.0332
Accuracy Degradation	N/A	4.50%
Energy Consumed (J)	100,200	50,490
Energy Saved	0	49,710 J (13 Wh)

2. COMMUNICATION REDUCTION



Timeline & Future Plan

Phase 1

APSM Filtering (Current)

- Developed the APSM filter for Gossip Learning.
- Successfully validated bandwidth reduction and accuracy on a 10-node edge network.

Phase 2

Extreme Scaling (Oct - Nov 2026)

- Stress-test the stability of the APSM protocol on massive, large-scale decentralized topologies (50-node and 100-node networks).

Phase 3

Agentic AI Integration (Jan - Feb 2027)

- Transition from passive forecasting to active management.
- Introduce Multi-Agent Reinforcement Learning (MARL). Deploy autonomous AI agents on edge nodes that ingest the APSM forecasts.

Phase 4

Autonomous Orchestration (Mar - Apr 2027)

- Finalize a fully autonomous, closed-loop edge network where AI agents negotiate and proactively offload tasks in real-time for improved resource utilization and resilience

References

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[2] A. Tundo, F. Filippini, F. Regonesi, M. Ciavotta, and M. Savi, “**Decentralized Edge Workload Forecasting With Gossip Learning,**” IEEE Transactions on Network and Service Management, vol. 22, no. 4, pp. 3016–3031, 2025, doi: 10.1109/TNSM.2025.3570450.

[3] G. Yao, H. Liu, and L. Dai, “**Multi-Agent Reinforcement Learning for Adaptive Resource Orchestration in Cloud-Native Clusters**,” in Proc. 2nd International Conference on Intelligent Computing and Data Analysis (ICDA), 2025, pp. 680–687, doi: 10.1145/3772726.3772833.

[4] E. Shahrad, R. Fonseca, I. Goiri, G. Chaudhry, P. Batum, J. Cooke, E. Laureano, C. Tresness, M. Russinovich, and R. Bianchini, "**Serverless in the Wild: Characterizing and Optimizing the Serverless Workload at a Large Cloud Provider,**" 2020 USENIX Annual Technical Conference (USENIX ATC 20), 2020, pp. 205-218.

THANK YOU

QUESTIONS?